

Data Visualization

- Importance of Data Visualization
 - Types of Data Visualization
 - Exploratory vs. Explanatory Visualization
 - Static vs. Interactive Visualization
 - Best Practices in Visualization
 - Overview of Popular Tools and Libraries:
 - Python Libraries: [Matplotlib](#), [Seaborn](#), [Plotly](#), [Altair](#), [Bokeh](#)
 - Other Tools: Tableau, Power BI
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2. Fundamentals of Visualization Design

- Understanding Your Audience
 - Choosing the Right Chart for the Data:
 - Bar, Line, Pie, Scatter, Histogram
 - Boxplots, Heatmaps, Treemaps, and Bubble Charts
 - Advanced Visuals: Sankey Diagrams, Network Graphs
 - Principles of Design:
 - Visual Hierarchy
 - Color Theory and Palette Selection
 - Effective Use of Annotations and Labels
 - Avoiding Misleading Visualizations
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3. Data Preparation for Visualization

- Data Cleaning and Wrangling:
 - Handling Missing Data
 - Transforming Data for Visualization
 - Aggregating and Summarizing Data
 - Filtering and Grouping Data
 - Reshaping Data (Pivoting, Melting)
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4. Visualization with Python Libraries

4.1 Matplotlib

- Introduction to Matplotlib
- Basic Plots: Line, Scatter, Bar
- Customizing Plots:
 - Titles, Labels, Legends
 - Colors and Styles
 - Gridlines and Subplots
- Advanced Features:
 - Dual Axis Plots
 - 3D Plots
 - Saving Plots as Images

4.2 Seaborn

- Overview of Seaborn
- Plotting Statistical Data:
 - Distribution Plots (Histograms, KDE, Rug)
 - Categorical Plots (Bar, Box, Violin, Swarm)
 - Matrix Plots (Heatmaps, Pairplots)
- Customizing Seaborn Visualizations
- Styling and Themes in Seaborn

4.3 Plotly and Cufflinks

- Creating Interactive Plots
- Line, Bar, and Scatter Plots
- Advanced Visualizations:
 - GeoPlots (Maps)
 - 3D Visualizations
 - Animations
- Exporting Interactive Visualizations

4.4 Bokeh

- Interactive Visualization with Bokeh
- Creating Basic Plots
- Adding Interactivity:
 - Tooltips
 - Widgets (Sliders, Dropdowns)
- Advanced Features:
 - Streaming Data
 - Building Dashboards

4.5 Altair

- Declarative Visualization with Altair
 - Creating Basic and Layered Plots
 - Encoding Data with Channels
 - Interactivity in Altair
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5. Specialized Visualization Techniques

5.1 Geospatial Visualization

- Working with Geospatial Data
- Choropleth Maps
- Point Maps
- Using Libraries:
 - `folium`, `geopandas`, `plotly`

5.2 Time Series Visualization

- Line Plots with Time Axes
- Annotating Events on Time Series
- Rolling Averages and Trends

5.3 Network Visualization

- Graphs and Networks
- Visualizing Social Networks
- Tools: `NetworkX`, `Plotly`

5.4 Dashboards

- Creating Dashboards with:
 - `Dash` (Python)
 - Tableau/Power BI
 - Integrating Multiple Plots and Data Sources
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6. Advanced Topics

6.1 Visualizing Big Data

- Handling Large Datasets
- Libraries for Big Data Visualization: [Dask](#), [Vaex](#)
- Aggregating Data for Effective Visualization

6.2 Animations in Visualization

- Adding Motion with [Matplotlib](#), [Plotly](#)
- Animated Time-Series Charts
- Using [celluloid](#) for Simple Animations

6.3 Custom Visualizations

- Building Custom Charts with [Matplotlib](#)
 - Customizing with [ggplot](#)-like Libraries ([plotnine](#))
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7. Visualization in Context

- Storytelling with Data
 - Embedding Visualizations in Reports and Presentations
 - Exporting Visualizations:
 - Static: PNG, PDF
 - Interactive: HTML, JavaScript Embeds
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8. Case Studies and Projects

Beginner Projects

- Visualizing a Dataset (e.g., Titanic, Iris)
- Sales Data Analysis (Bar, Line Charts)

Intermediate Projects

- Exploring Correlations with Heatmaps
- Time-Series Analysis (Stock Market, Weather Data)

Advanced Projects

- Creating a Geospatial Dashboard

- Building Interactive Multi-Page Dashboards with [Dash](#)
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9. Tools and Ecosystem

- Python IDEs for Visualization ([Jupyter](#), [VSCode](#))
- Integrating with Other Tools:
 - Pandas for Data Wrangling
 - SQL for Data Queries
 - APIs for Real-Time Data

Matplotlib

- What is Matplotlib?
 - Key Features and Applications
 - Installing Matplotlib
 - Anatomy of a Matplotlib Plot
 - Figure, Axes, and Axis
 - Comparing Matplotlib with Other Libraries (e.g., Seaborn, Plotly)
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2. Basics of Matplotlib

2.1 Simple Plots

- Creating a Basic Line Plot
- Adding Titles, Labels, and Legends
- Customizing Plot Styles and Colors
- Displaying and Saving Plots ([plt.show](#), [plt.savefig](#))

2.2 Using Pyplot

- Overview of the Pyplot Interface
- Commonly Used Pyplot Functions ([plot](#), [scatter](#), [bar](#), etc.)
- Managing Figures and Axes

3. Customizing Plots

- Titles, Labels, and Legends:
 - Adding and Styling
 - Controlling Position
 - Customizing Lines:
 - Line Style (`linestyle`, `linewidth`)
 - Markers and Colors
 - Axes Customization:
 - Setting Axis Limits (`xlim`, `ylim`)
 - Logarithmic Scales
 - Formatting Ticks and Grids
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4. Types of Plots

4.1 Line Plots

- Creating Line Graphs
- Multiple Lines on a Single Plot
- Styling Lines (Dashes, Widths)

4.2 Scatter Plots

- Plotting Points with `scatter`
- Customizing Marker Size, Color, and Transparency

4.3 Bar Plots

- Vertical and Horizontal Bars
- Grouped and Stacked Bar Charts
- Adding Error Bars

4.4 Histograms

- Creating Histograms
- Customizing Bins
- Overlapping Histograms

4.5 Pie Charts

- Creating Pie Charts
- Customizing Labels and Exploding Slices

4.6 Box Plots

- Creating and Customizing Box Plots
- Adding Custom Notches and Whiskers

4.7 Heatmaps

- Creating Heatmaps with `imshow`
- Customizing Color Maps and Scales

4.8 Other Plot Types

- Area Plots
 - Step Plots
 - Error Bar Plots
 - Violin Plots
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5. Subplots and Layouts

- Creating Multiple Plots in One Figure (`subplot`, `subplots`)
 - Adjusting Layout with `tight_layout` and `GridSpec`
 - Sharing Axes Between Subplots
 - Overlapping and Inset Plots
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6. Advanced Customization

- Adding Annotations and Text
 - Working with Colors:
 - Color Maps
 - RGB and Hex Values
 - Working with Styles and Themes (`plt.style.use`)
 - Customizing Legends:
 - Positioning
 - Multiple Legends
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7. Working with Dates and Times

- Plotting Time-Series Data
 - Formatting Date Axes
 - Handling Timezones
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8. 3D Plotting

- Setting Up 3D Axes
 - Creating 3D Line and Scatter Plots
 - Surface Plots and Wireframes
 - Contour Plots in 3D
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9. Interactive Features

- Adding Widgets (e.g., Sliders, Buttons)
 - Interactive Backends (`nbagg`, `Qt5Agg`)
 - Zooming and Panning in Interactive Mode
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10. Animations in Matplotlib

- Creating Basic Animations (`FuncAnimation`)
 - Customizing Animation Frames
 - Saving Animations as GIF or Video
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11. Working with Images

- Displaying Images with `imshow`
 - Applying Filters and Transformations
 - Working with Color Maps and Pixel Data
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12. Performance Optimization

- Reducing Rendering Time for Large Datasets
 - Using `Agg` Backend for Faster Plots
 - Optimizing with `blit` for Animations
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13. Integration with Other Libraries

- Combining Matplotlib with Pandas:
 - Plotting Directly from DataFrames
 - Using Matplotlib with Seaborn
 - Creating Dashboards with Matplotlib in `Dash` or `Streamlit`
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14. File I/O

- Saving Plots:
 - Supported Formats (PNG, JPG, PDF, SVG)
 - DPI and Resolution Settings
 - Exporting to Vector Graphics
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15. Case Studies and Real-World Applications

Beginner Projects

- Visualizing Simple Linear Trends
- Creating Basic Bar and Pie Charts

Intermediate Projects

- Plotting Time-Series Data
- Comparing Data Distributions with Histograms

Advanced Projects

- Building Multi-Plot Figures for Reports
- Animating a Dynamic System (e.g., Pendulum Motion)

16. Advanced Topics

- Customizing Ticks and Grids with `matplotlib.ticker`
 - Understanding and Using Transforms
 - Creating Custom Plot Types
 - Building Custom Themes and Styles
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Seaborn

- What is Seaborn?
 - Key Features and Advantages
 - Installing Seaborn
 - Comparison with Matplotlib and Other Libraries
 - Anatomy of a Seaborn Plot
 - Loading Built-in Datasets (`load_dataset`)
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2. Basics of Seaborn

- Overview of Seaborn's Functional Interface
 - The Role of Pandas DataFrames in Seaborn
 - Understanding the Concept of Figure Aesthetics
 - Customizing Plots with Matplotlib Context
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3. Seaborn Themes and Styles

- Using Predefined Themes (`darkgrid`, `whitegrid`, `dark`, `white`, `ticks`)
 - Changing Plot Aesthetics with `set_theme` and `set_style`
 - Customizing Context for Different Scenarios (`paper`, `notebook`, `talk`, `poster`)
 - Modifying Font Sizes and Line Widths (`set_context`)
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4. Univariate Data Visualization

4.1 Distribution Plots

- Creating Histograms (`histplot`)
 - Kernel Density Estimation (`kdeplot`)
 - Combining Histograms and KDE
 - Rug Plots (`rugplot`) for Small Data Visualizations
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5. Bivariate Data Visualization

5.1 Scatter Plots

- Creating Basic Scatter Plots (`scatterplot`)
- Customizing Marker Size, Color, and Transparency
- Adding Regression Lines with `regplot`

5.2 Line Plots

- Creating Line Charts (`lineplot`)
- Working with Time-Series Data
- Customizing Styles and Line Widths

5.3 Hexbin and 2D KDE Plots

- Creating Hexbin Plots (`jointplot` with `kind="hex"`)
 - Visualizing 2D KDE (`kdeplot`, `jointplot` with `kind="kde"`)
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6. Categorical Data Visualization

6.1 Bar Plots

- Creating Bar Charts (`barplot`)
- Aggregating Data with `estimator`
- Customizing Error Bars

6.2 Count Plots

- Plotting Frequency Counts (`countplot`)
- Customizing Colors and Orientations

6.3 Box and Violin Plots

- Creating Box Plots (`boxplot`)
- Customizing Whiskers, Notches, and Outliers
- Violin Plots for Distribution Visualization (`violinplot`)
- Splitting Violin Plots

6.4 Strip and Swarm Plots

- Creating Jittered Data Plots (`stripplot`)
 - Overcoming Overlap with Swarm Plots (`swarmplot`)
 - Combining Plots (e.g., Box + Swarm)
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7. Matrix Plots

7.1 Heatmaps

- Creating Heatmaps with `heatmap`
- Customizing Color Scales, Annotations, and Labels
- Visualizing Correlation Matrices

7.2 Cluster Maps

- Visualizing Hierarchical Clustering with `clustermap`
 - Customizing Row and Column Linkages
 - Using Different Distance Metrics
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8. Pairwise Relationships

8.1 Pair Plots

- Creating Pairwise Relationships with `pairplot`
- Visualizing Multi-Dimensional Data
- Adding KDE or Histograms to Pair Plots

8.2 Joint Plots

- Creating Combined Scatter and KDE Plots (`jointplot`)
 - Customizing Plot Types (`scatter`, `kde`, `reg`, `hex`)
 - Adding Marginal Histograms or KDEs
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9. Customizing Seaborn Plots

- Adding Titles, Labels, and Legends
 - Changing Axes Limits and Ticks
 - Customizing Colors and Palettes (`color_palette`, `cubehelix_palette`)
 - Working with Custom Color Maps
 - Adding Annotations and Text
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10. Working with DataFrames

- Using `hue`, `size`, and `style` Parameters for Grouping
 - Faceted Plots with `FacetGrid`
 - Creating Multi-Panel Plots:
 - `catplot`
 - `relplot`
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11. Advanced Topics

11.1 Statistical Visualization

- Visualizing Regression with `lmplot`
- Plotting Residuals with `residplot`
- Logistic Regression and Higher-Order Models

11.2 Plotting Large Datasets

- Working with Subsampled Data
- Using `hue_order` and `size_order`

12. Combining Seaborn and Matplotlib

- Customizing Subplots with `FacetGrid.map`
 - Using Matplotlib's `Axes` for Advanced Customizations
 - Integrating Seaborn Plots in Matplotlib Subplots
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13. Animations and Interactive Plots

- Creating Animations with Seaborn and `FuncAnimation`
 - Exporting Interactive Visualizations
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14. File I/O

- Saving Seaborn Visualizations to File (`savefig`)
 - Exporting High-Resolution Images for Publications
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15. Projects and Case Studies

Beginner Projects

- Visualizing a Single Variable with Distribution Plots
- Exploring Relationships in Built-in Datasets (e.g., Iris, Titanic)

Intermediate Projects

- Creating a Correlation Heatmap for a DataFrame
- Building Multi-Panel Categorical Visualizations

Advanced Projects

- Analyzing Multi-Dimensional Data with Pair Plots
 - Building a Dashboard with Matplotlib and Seaborn
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16. Real-World Applications

- Statistical Analysis and Data Exploration
- Visualizing Machine Learning Results
- Data Storytelling with Seaborn